

1.5a

Sleep

Consciousness

What is Consciousness?

in Ψ , Consciousness is a fundamental, yet hard-to-define, concept

How has our understanding of consciousness changed?

Initially, Ψ was defined as the study of consciousness,

During first $\frac{1}{2}$ of 1900s, many started defining Ψ as the study of behavior

In the 1960s, Ψ started reconsidering consciousness. Neuro science advances linked brain activity to sleeping, dreaming & other mental states.

Research into altering consciousness with drugs, meditation, "hypnosis". Importance of cognition $\therefore$ mental processes

Today, many Psychologists define consciousness as our subjective awareness of ourself & our environment.

Conscious awareness helps to

- ★ Make sense of life (sensations, emotions, choices)
- ★ Goal-set; plan; most thoughts focus on present, future
- ★ focus attention while learning behavior

States of Consciousness

Over time we flip between different states of consciousness: (normal waking awareness, other altered states)

Some states occur spontaneously:

Daydreaming, Flow, Dreaming

Some are physiologically induced:

Hallucinations, Orgasm, Food or Oxygen Starvation

Some are psychologically induced:

Sensory Deprivation, "Hypnosis", Meditation

How does the brain make the mind? : Cognitive Neuroscience

Cognitive Neuroscience: the interdisciplinary study of the brain activity linked with our mental-processes.

skilled soccer players were asked to think about creative vs simple soccer moves.

Scans showed that thinking about creative moves produced the most coordinated brain activity.

noncommunicative car accident victim
no outward signs of conscious awareness
researchers asked her to imagine playing tennis
fMRI scans showed activity in areas controlling arm/leg movement
Even in a motionless, noncommunicative body - the mind may be active!
More studies shown that 15-30% of unresponsive patients may
be experiencing meaningful conscious awareness

Cognitive neuroscientists can, in a way, read minds.
Based on cortical activation, they could tell, for
example, which of 10 similar objects (ex. Hammer, Nail) you
were viewing

If a stimulus activates enough brain-wide coordinated
activity, it crosses a threshold for consciousness

A weaker stimulus (ex. a word flashed too briefly to be
consciously perceived.) may trigger localized activity that
quickly fades.

Stronger activity may engage other brain areas.
Such reverberating activity is a sign of
conscious awareness. This can provide another indication
of awareness in unresponsive patients.

How this activity produces awareness is a mystery.

How does our subconscious work?

When we look at a bird flying, we are consciously aware of it. But not of our subprocessing of the bird's color, form, movement & distance.

Thinking, knowing, remembering & communicating operate on 2 independent levels:

a conscious, deliberate "high road"

an unconscious, automatic, "low road"

Together, they create dual processing.

We know more than we know we know

The human brain converts conscious thought into unconscious knowledge.

D.F. suffered a brain injury, she could no longer recognize & discriminate objects.

She had blind sight: she acted like she could see

Asked to slip a postcard into a vertical or horizontal slot, she could do so w/o error.

Asked to approximate the width of a block, she was at a loss but could grip it with the right finger-thumb distance.

D.F. could not consciously recognize objects, but could subconsciously understand their characteristics

Our vision is a dual-processing system
Visual Perception track → to think about the world
Visual Moment track → guides moment to moment actions.

A man lost left-visual-cortex ∴ he's blind to objects & faces on the right side of his vision.

But he could recognize emotions on these faces

80-90% of what we do is unconscious

Subconscious biases can overpower conscious intentions

The unconscious parallel processing is faster than conscious sequential processing, but both are essential

Parallel processing does routine tasks
Sequential processing solves new problems.

MCA:

1. d 2. b 3. a 4. c 5. b